A Product Analytics Project

Can we replace a 5-second video limit with a smarter rule?

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Swipe To See The Breakdown →

The Context

current reconstruction pipeline

Source clip → First frame & description → Generative model →Reconstructed video

When reconstruction fails, the model can:

- invent objects
- mutate the environment
- morph subjects
- turn text into nonsense

Today's Rule

≤ 5 seconds → automated
> 5 seconds → manual

But duration is only a proxy

we want to automate clips up to 7 seconds because the 5–7 second band represents costly manual routing

THE QUESTION:

Can we quantify the visual characteristics associated with reconstruction risk and use them to decide which clips are safe to automate?

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FROM “VISUAL COMPLEXITY” → TO MEASURABLE SIGNALS

Operators described risky clips as those with:

① Overall frame-to-frame change
How much does the image change across the clip?

② Peak frame-to-frame change
What is the largest instantaneous visual change?

③ Global visual/compositional change
How much does the overall image distribution change?

The three Signals I Measured

Mean Frame Difference

Max Frame Difference

Max Histogram Difference

Why only three?

I wanted a small, interpretable model rather than a collection of loosely related features.

WHAT I FOUND IN 100 CLIPS

The three signals captured different kinds of visual change.

Mean ↔ Max Frame → **r = 0.67**

Related, but not identical

Max Frame ↔ Max Histogram → **r = 0.72**

Strong relationship, but each captures a different aspect of change

Mean Frame ↔ Max Histogram → **r = 0.35**

Relatively distinct signals.

Why keep all three?

Mean frame difference captures overall change.
Max frame difference captures sudden peaks of change.
Max histogram difference captures global compositional change.

No single metric captures every kind of visual change

So I didn't choose one “best” metric

I combined complementary signals.

FROM 3 SIGNALS → 1 RISK SCORE

We show:

Mean Frame Difference
Max Frame Difference
Max Histogram Difference

↓

normalize each → 40% Mean + 30% Max Frame + 30% Max Histogram

↓

Eligible / Exclude ← RISK SCORE

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WHAT DOES THE SCORE LOOK LIKE IN PRACTICE?

Example: Block Blast.mp4
(A video from the File)

Observed signals

Signal	Normalized
Mean Frame Difference	0.95
Max Frame Difference	0.79
Max Histogram Difference	0.67

RISK SCORE = 81.8 ⟶ EXCLUDE

High visual-change score → route away from automated reconstruction.

FROM SCORE → ROUTING DECISION

Across 100 clips:
68 → ELIGIBLE
32 → EXCLUDE

	Manual: ELIGIBLE	Manual: EXCLUDE
Model: ELIGIBLE	54	18
Model: EXCLUDE	14	14

- **14 clips that I manually considered eligible were excluded by the model.**
- **18 clips that I manually considered excluded were still marked eligible.**

The score doesn't eliminate judgment — it concentrates it where the signals are ambiguous.

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